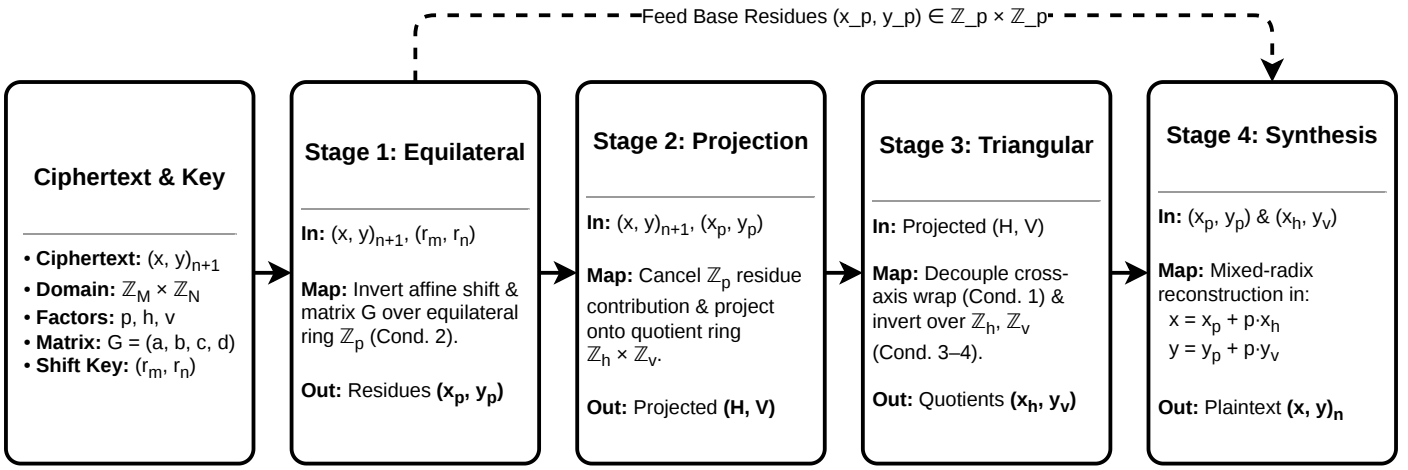
Feed Base Residues (x_p, y_p) ∈ ℤ_p × ℤ_p
Ciphertext & Key
• Ciphertext: (x, y)n+1
• Domain: ℤM × ℤN
• Factors: p, h, v
• Matrix: G = (a, b, c, d)
• Shift Key: (rm, rn)
Stage 1: Equilateral
In: (x, y)n+1, (rm, rn)
Map: Invert affine shift & matrix G over equilateral ring ℤp (Cond. 2).
Out: Residues (xp, yp)
Stage 2: Projection
In: (x, y)n+1, (xp, yp)
Map: Cancel ℤp residue contribution & project onto quotient ring ℤh × ℤv.
Out: Projected (H, V)
Stage 3: Triangular
In: Projected (H, V)
Map: Decouple cross-axis wrap (Cond. 1) & invert over ℤh, ℤv (Cond. 3–4).
Out: Quotients (xh, yv)
Stage 4: Synthesis
In: (xp, yp) & (xh, yv)
Map: Mixed-radix reconstruction in:
x = xp + p·xh
y = yp + p·yv
Out: Plaintext (x, y)n